

COMPUTER VISION • PHONE CAMERA LAB 3

From light to images

Exposure, image enhancement, perspective, and focus

Your task: use your phone photographs to investigate image formation and processing. Identify what information comes from one image and what requires additional captures. Build the results in Python; automatic phone outputs are not substitutes.

Format: two suggested 90-minute sessions; pairs sharing one phone. Complete A–D. Use library functions except where implementation is explicitly required. Finish figures and discussion after class as needed.

Meeting	Suggested allocation	Main result
Session 1	Setup 10 min; A 25 min; B 35 min; capture C/D 20 min	Perspective measurements, histograms, and edge maps
Session 2	C 40 min; D 35 min; review 15 min	Exposure fusion/HDR and an all-in-focus composite

Before you start

Bring a phone, laptop, ruler or tape measure, two identical cards, three textured objects, and a stable phone support. Books can replace a tripod. Use Python with NumPy, OpenCV, and Matplotlib; a local notebook or an existing course environment is sufficient. No depth sensor, second camera, paid app, or extra lens is required.

Use ordinary rear-camera photographs. Keep the same lens, zoom, resolution, and orientation within each sequence. Turn off flash, filters, and synthetic background blur. Disable automatic HDR/night processing where possible; otherwise record that it remains active. JPEG images are acceptable, but do not describe them as unprocessed sensor data. Preserve original files and make processing copies rather than sending images through a service that resizes them.

Start `capture_log.csv`: record filename, scene, phone/app, selected camera, image dimensions, exposure time, ISO, focus setting, and enabled processing. Write **unknown** for unavailable settings. Record actual capture settings, not just the intended settings. Photograph objects rather than identifiable people or private information; never aim at the sun.

Keep the three capture experiments separate

Experiment	Deliberately change	Keep fixed
Perspective	Camera-to-scene distance	Objects, lens, and zoom
Exposure bracket	Exposure time, or exposure compensation	Camera position, scene, focus, and preferably ISO/white balance
Focus stack	Focus setting	Camera position, scene, exposure, lens, and zoom

Phone limitations: exposure compensation is enough for the exposure-fusion route. Tap-to-focus is acceptable when manual focus is unavailable. A genuinely fixed-focus camera cannot produce a focus sweep: document your attempt, capture the other tasks yourself, and use an identified classmate's stack for D. This carries no hardware penalty.

Reading connection: pinhole geometry: lec4-1.pdf, pp. 2, 8–11, 18, 30; derivatives/features: lec3.pdf, pp. 10–28. Exposure and focus extend the classroom discussion, not the uploaded geometry slides.

SESSION 1 • SINGLE-IMAGE MEASUREMENTS

A. Distance, perspective, and field of view

Capture. Place two equal-height cards at different depths, laterally separated so both remain visible. Keep both cards upright and facing the camera. Start with the nearer card about 0.7 m away and the other about 0.3 m behind it; adapt these distances to your room. Take one photograph. Move the phone backwards approximately 2 m without moving the cards or changing zoom, then take a second photograph. Measure the two card depths from each camera position.

Measure and explain. Measure each card's pixel height in both images. Use the positive virtual-image-plane convention from lec4-1.pdf, p. 30:

$$h = \frac{f_y H}{Z}, \quad \frac{h_{\text{near}}}{h_{\text{far}}} = \frac{Z_{\text{far}}}{Z_{\text{near}}}.$$

Here H is physical card height, h is pixel height, and f_y is focal length in pixels. Produce a four-row measurement table: camera position, card, measured depth, and pixel height. Compare the measured height ratio with the predicted ratio for each photograph.

Crop and enlarge the distant photograph until the near card resembles its size in the close photograph.

Does the relative size of the two cards become the same? Use your measurements to explain why cropping is not a change of viewpoint.

A weak-perspective approximation replaces the varying depth Z by a representative depth Z_0 . Discuss whether your distant capture better satisfies $|Z - Z_0|/Z_0 \ll 1$. Being far away alone is not the criterion; the scene's depth variation must be small relative to its distance. Distinguish this perspective effect from curved straight lines caused by lens distortion.

Optional lens comparison: from a fixed phone position, compare the main and ultra-wide cameras, where available. Record the different fields of view. Do not treat this as a pure focal-length experiment: switching physical cameras also shifts the optical center. A digital crop is not optical zoom.

B. Histograms, enhancement, and edges

Photograph a low-contrast textured scene and a scene containing both bright and dark areas. On grayscale versions, compute a 256-bin histogram and cumulative distribution. **Implement the histogram/CDF lookup-table mapping yourself**, then compare it with `cv.equalizeHist`. Handle a constant-valued image without division by zero. Compare global equalization with `cv.createCLAHE(clipLimit=2.0, tileGridSize=(8,8))`. [W1]

For one image, apply Gaussian smoothing with standard deviations 1 and 3 pixels, followed by Sobel gradients. Display the gradient magnitude $G = \sqrt{G_x^2 + G_y^2}$ and a Canny edge map. Compute derivatives in floating point before taking their magnitude. Try two Canny threshold pairs while keeping other settings unchanged. [W2]

Record: original/enhanced images with histograms, smoothing/edge comparisons, and two short observations. Which details become easier to see? Which noise or weak edges become worse? Explain why remapping identical clipped pixel values cannot recover the scene differences that were lost there. Equalization is a CDF remapping, not merely linear contrast stretching.

SESSION 2 • EXPOSURE SEQUENCES

C. Build an exposure-fusion or HDR result

Capture genuinely different exposures

Choose a stationary scene containing a bright window and a darker indoor object with printed detail. Support the phone. Take **at least three separate photographs**: dark, middle, and bright. Keep the composition unchanged. Do not create the sequence by digitally brightening one photograph.

With manual shutter control, fix ISO, focus, lens, and white balance, then try exposure times $t_0/4$, t_0 , and $4t_0$. These are -2 , 0 , and $+2$ stops relative to t_0 at fixed aperture and ISO. Record the actual exposure times in seconds. With exposure-compensation controls only, aim for a similarly varied bracket, but do not assume that the phone changed shutter time alone.

Inspect the images before processing. Identify one shadow patch and one highlight patch whose readable detail changes across exposures. Repeat the capture when every image is nearly identical or when all images lose the same important detail. Report automatic processing that you could not disable.

Process one of the following routes

Route C1 — all phones: exposure fusion. Align small camera shifts if necessary, crop to common valid coverage, and use `cv.createMergeMertens().process(images)`. Also compute a simple average of the aligned images in floating point. Compare both with the middle exposure and with equalization of that middle exposure. Mertens fusion produces a display-oriented image; it does not recover a calibrated radiance map and does not require exposure times. [W3]

Route C2 — known shutter times and fixed ISO: approximate radiometric HDR. Estimate a camera response with `cv.createCalibrateDebevec`, merge with `cv.createMergeDebevec`, and tone-map the floating-point result, for example with `cv.createTonemapReinhard`. Supply the exposure times in the same order as the images. Preserve the floating-point result separately from the display image. [W3, W4]

For C2, discuss the fixed-response assumption. Per-frame local tone mapping, denoising, or hidden fusion in processed phone files may violate it. A successful-looking merge is not proof that its values are physically calibrated. Choose C1 when the capture settings or processing are uncertain; **both routes can earn full credit**.

Evidence and questions

Submit an exposure contact sheet, capture settings, and comparisons of the same highlight and shadow patches. Compute the fraction of grayscale pixels at or below 5 and at or above 255 in each input; label this an **exported-image near-clipping indicator**, not a sensor dynamic-range measurement.

Explain: Which input supplies useful detail in each patch? Why is averaging not generally the best merge? How does multi-exposure capture differ from equalizing one image? For C2, distinguish radiance reconstruction from tone mapping; for C1, explain why your output should be called exposure fusion.

Failure test: deliberately move one object between a second set of exposures. Compare the result with your stationary set and identify ghosting or inconsistent content. Do not use the moving set as your main measurement.

SESSION 2 • FOCUS SEQUENCES

D. All-in-focus imaging and depth from focus

Capture a focus stack without moving the phone

Arrange three textured objects at near, middle, and far distances. A printed label, textured box, and book cover work well. Keep the phone on its support and stay within the chosen camera's focusing range. Avoid Portrait mode and automatic switching to another lens. Keep lighting, exposure, ISO, and white balance as consistent as the phone allows.

Take **5–10 images**, changing only the focus setting from near to far. With tap-to-focus, capture at least three distinct settings and record the target used for each. Verify that different objects actually become sharper in different images. When everything is sharp in every frame, move the scene closer during setup, increase the separation between its depth layers, and then repeat the whole stack from a fixed position.

Record object distances for interpreting the result. Do not move the phone between stack images: that introduces viewpoint and scale changes, making this a different experiment. Small magnification changes caused by focusing may still occur; inspect alignment and use a common crop after registration if necessary.

Implement local sharpness selection

Convert each aligned image I_k to grayscale floating point in the same intensity range. Compute a local gradient-energy score using Sobel derivatives and an averaging window, initially 9×9 pixels:

$$S_k(x, y) = \text{mean}_{(u,v) \in W(x,y)} (G_{x,k}(u, v)^2 + G_{y,k}(u, v)^2).$$

For each pixel, choose the image with the largest local score and copy its color pixel to the composite:

$$j(x, y) = \operatorname{argmax}_k S_k(x, y), \quad I_{\text{fused}}(x, y) = I_{j(x,y)}(x, y).$$

Implement this selection yourself. OpenCV may supply filtering and registration. Do not choose one globally sharpest image: different regions need different source frames. Compare two window sizes and explain the trade-off near boundaries. A simple selection map may produce seams; inspect these rather than hiding them. Focal-stack compositing is the broader idea behind this exercise. [W5]

Interpret the result honestly

Display the all-in-focus composite and the selected **focus-index map**, with a legend matching indices to capture order. Plot mean sharpness versus focus index for a small textured patch on each object. Compare the winning indices with the known near/middle/far ordering.

Mark unreliable pixels where the maximum score is small or the two best scores are too similar. State your chosen thresholds. Textureless surfaces, exposure changes, noise, and misregistration can all corrupt this particular sharpness-selection experiment.

A focus index is not a distance in meters. Metric depth requires a calibrated relationship between focus setting and object distance, plus reliable focus measurements. For example, Apple explicitly states that its `lensPosition` value is not an exact physical distance. [W6]

Submit: representative focused inputs, three sharpness curves, the composite, the index map and invalid-region mask, and one failure explanation. A documented fixed-focus limitation plus analysis of an identified classmate's stack is acceptable; keep that data separate from your own captures.

SUBMISSION • LAB 1

What to submit

Submit one folder named Lab1_GroupID containing original captures, capture_log.csv, a runnable notebook or Python script, exported results, and a **report of no more than four pages**. Put additional contact sheets in an appendix. Include team member names and who captured, programmed, and checked each part.

Your report should answer the questions beside each task using your own images and measurements. Use the correct labels: **display image**, **approximate HDR**, **focus index**, and **unknown setting** where appropriate. A convincing explanation of a failed experiment is more valuable than an attractive output with an incorrect claim.

Assessment: 100 points

Criterion	Points	What earns credit
Capture control and reproducibility	15	Own data, useful logs, consistent sequences, honest limitations
A. Perspective experiment	20	Quantitative ratios and correct distinction between distance, cropping, and lens effects
B. Enhancement and edges	20	Correct CDF implementation, comparisons, and interpretation
C. Multi-exposure processing	25	Correct route, regional evidence, and analysis of missing detail/ghosting
D. Focus experiment	20	Local selection, evidence of focus variation, and appropriate uncertainty/units

Hardware capability does not increase or reduce the available marks. A justified focus fallback is assessed through your documented limitation, implementation, and analysis. Automatic phone outputs may be additional comparisons, not substitutes for the required processing.

Optional investigations

Choose one only after completing the core: compare motion blur at two recorded shutter times; investigate ISO while recording exposure time and avoiding claims that gain collects more photons; change white balance on a neutral scene; or show how a small Harris-corner experiment responds to blur. Change one variable at a time while controlling the others. Do not attribute a specific processing pipeline to a manufacturer without evidence.

Implementation references

[W1] OpenCV: [Equalization/CLAHE](#). [W2] OpenCV: [Gradients](#). [W3] OpenCV: [HDR and exposure fusion](#). [W4] Debevec & Malik (1997): [HDR radiance maps](#). [W5] Jacobs et al. (2012): [Focal-stack compositing](#). [W6] Apple: [lensPosition](#).